

Talha Jubayer Asif

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Career Objective

Final-year Computer Science & Engineering student at AIUB with a strong foundation in computer networks, network security, and Linux, seeking a networking-focused internship to apply hands-on knowledge of TCP/IP, network traffic analysis, network reconnaissance, and security fundamentals. Experienced with Nmap, Wireshark, Kali Linux, and VirtualBox through academic cybersecurity projects, with additional exposure to machine learning-based intrusion detection and web technologies. Eager to contribute to network operations and security while developing practical experience in enterprise networking environments.

Education

Bachelor of Science in Computer Science & Engineering *Sept 2022 - Jan 2027 (Expected)*
American International University-Bangladesh (AIUB) — Dhaka, Bangladesh

- CGPA: **3.41 / 4.00** Total Credits: **148**
- Faculty of Science and Technology (FST)
- Completed 142 of 148 credits (remaining: thesis and internship)

Higher Secondary Certificate (HSC) 2021
Govt. Shahpur Madhugram College, Board — Jashore

- GPA: **5.00 / 5.00** Group: Science

Major Courses

- Network Security
- Human Computer Interaction
- Computer Vision & Pattern Recognition
- Machine Learning
- Management Information System

Relevant Core Coursework

- Artificial Intelligence & Expert Systems
- Algorithms & Data Structures
- Computer Networks & Data Communication
- Operating Systems
- Software Engineering
- Database Management Systems
- Object-Oriented Programming (OOP 1 & 2)
- Web Technologies
- Computer Organization & Architecture
- Computer Graphics
- Compiler Design
- Microprocessor & Embedded Systems
- Theory of Computation
- Digital Logic & Circuits

Technical Skills

Network & Security	TCP/IP, Nmap, Wireshark, John the Ripper, SQLi, Cryptography, Firewalls, VPN Concepts
ML / AI Frameworks	TensorFlow, Scikit-learn, OpenCV, NumPy, Pandas
Computer Vision	Image Classification, Object Detection, Pattern Recognition
Web Technologies	HTML5, CSS3, JavaScript, PHP, MySQL
Database	MySQL, Oracle SQL
Tools & Platforms	Git, GitHub, Linux, VS Code, Google Colab
Others	OOP, SDLC, UML, Agile Basics, UI/UX Principles (HCI)

Academic Projects

Basic Penetration Testing Using Kali Linux & DVWA Summer 2026
Course: Network Security

- Configured an isolated VirtualBox NAT lab and performed Nmap SYN scans for network reconnaissance, service discovery, and vulnerability identification.

- Used Wireshark to analyze HTTP traffic and inspect session data; performed controlled password and hash-security testing using Hydra and John the Ripper.
- Documented identified vulnerabilities and proposed remediation including TLS, secure password hashing, and patch management.

Robust and Explainable Machine Learning for Multi-Stage Attack Detection in Smart Home Networks

Ongoing, Summer 2026

Conference Research Project — Cybersecurity

- Developed an ML-based intrusion detection framework for multi-stage attacks using a Smart Home Intrusion Detection Dataset.
- Compared Logistic Regression, Random Forest, XGBoost, and MLP using F1-score, PR-AUC, ROC-AUC, and False Positive Rate.
- Evaluated generalization using unseen-attack scenarios and applied feature selection and SHAP-based explainability.

Attention-Enhanced Agricultural Image Classification Using EfficientNet-B3 with CBAM and XAI

Summer 2026

Course: Computer Vision & Pattern Recognition

- Built a 77-class agricultural image classifier using pretrained EfficientNet-B3 and CBAM attention on 5,405 images.
- Applied augmentation, class-weighted label-smoothing loss, AdamW optimization, and cosine learning-rate scheduling.
- Achieved 87.32% test accuracy, 86.43% Macro F1, and 86.74% Weighted F1; evaluated model variants through ablation studies and XAI visualization.

Simple Quiz Application — GitHub Repository

Summer 2026

Course: Web Technologies

- Developed an MVC-based quiz web application using PHP, MySQL, HTML, CSS, and structured routing with `.htaccess`.
- Used MySQL for database management and applied modular software design.
- Used Git/GitHub for source-code management.

Co-curricular Activities & Achievements

- Member of The Duke Of Edinburgh's International Award AIUB (DEA AIUB)

Languages

Bangla Native / Mother Tongue

English Professional Working Proficiency